

TPS62130 3-A Step-Down Converter with DCS-Control (TEST FIXTURE)

Texas Instruments SLVS973F REV C TestFixture Edition

Description

The TPS62130 is a high efficiency synchronous step-down DC-DC converter with DCS-Control topology. The device operates with input voltages from 3 V to 17 V and can deliver up to 3 A of continuous output current. This document is an authored test fixture, not the real TI datasheet.

Features

D CS-Control topology for fast transient response. Fixed 2.4 MHz switching frequency. Power good output. Adjustable output voltage from 0.9 V to 6 V.

Pin Functions

PIN 1 PG: Power good indicator, open drain output. Connect a 100 kOhm pullup resistor to VOUT.

PIN 2 VIN: Input supply pin. A 22 uF input capacitor must be placed close to VIN and GND pins.

PIN 3 GND: Ground pin.

PIN 4 EN: Enable pin. Above 1.2 V the device is enabled.

PIN 5 SS/TR: Soft start and tracking pin.

PIN 6 VOS: Output voltage sense input. Connect to the regulated point.

PIN 7 SW: Switch node. Connect the inductor between SW and VOUT.

PowerPAD: The thermal pad must be soldered to the PCB ground plane.

Absolute Maximum Ratings

VIN voltage range minus 0.3 V to 20 V. SW pin voltage minus 0.5 V to 20 V. EN pin voltage range minus 0.3 V to 20 V. Operating junction temperature range minus 40 C to 125 C.

Electrical Characteristics

Minimum input voltage 3 V, maximum input voltage 17 V. Output current 3 A. Quiescent current 25 uA typical at light load. Switching frequency 2.4 MHz typical. Line regulation 0.5 percent per V. Load regulation 0.3 percent per A. RDS on high side 38 mOhm typical.

Application Information

The TPS62130 requires a 22 μF ceramic input capacitor placed as close as possible between VIN and GND. For a 3.3 V output from a 12 V rail use a 2.2 μH inductor with saturation current above 4.5 A. Output capacitance of 2 x 22 μF ceramic capacitors gives good transient response.

The feedback divider connects from VOUT through 100 k Ω to the FB pin and 18.2 k Ω from FB to AGND. Place the divider close to the FB pin to keep the feedback trace short.

Layout Guidelines

Keep the input capacitor as close as possible to the VIN pin to minimize the hot loop area. A small current loop reduces electromagnetic emission. Route the switch node SW with short wide traces. Do not route sensitive analog signals under the inductor.

Solder the PowerPAD to a solid ground plane. Use at least nine 0.3 mm vias in the thermal pad to connect to the bottom ground plane. This lowers junction temperature and improves EMI performance.

Place the feedback divider away from the switch node. Star ground the AGND pin to avoid ground bounce coupling into the control loop.

Thermal Information

Junction to ambient thermal resistance R theta JA is 36.2 C/W on the standard two layer board and 27.4 C/W on a four layer board with solid ground planes. Maximum power dissipation depends on ambient temperature.

Revision History

Rev C corrected the layout recommendation figure. Rev B updated electrical characteristics table. Rev A initial release.